EC-390: Digital Signal Processing

LECTURE NO 3: *Analog to Digital Conversion*

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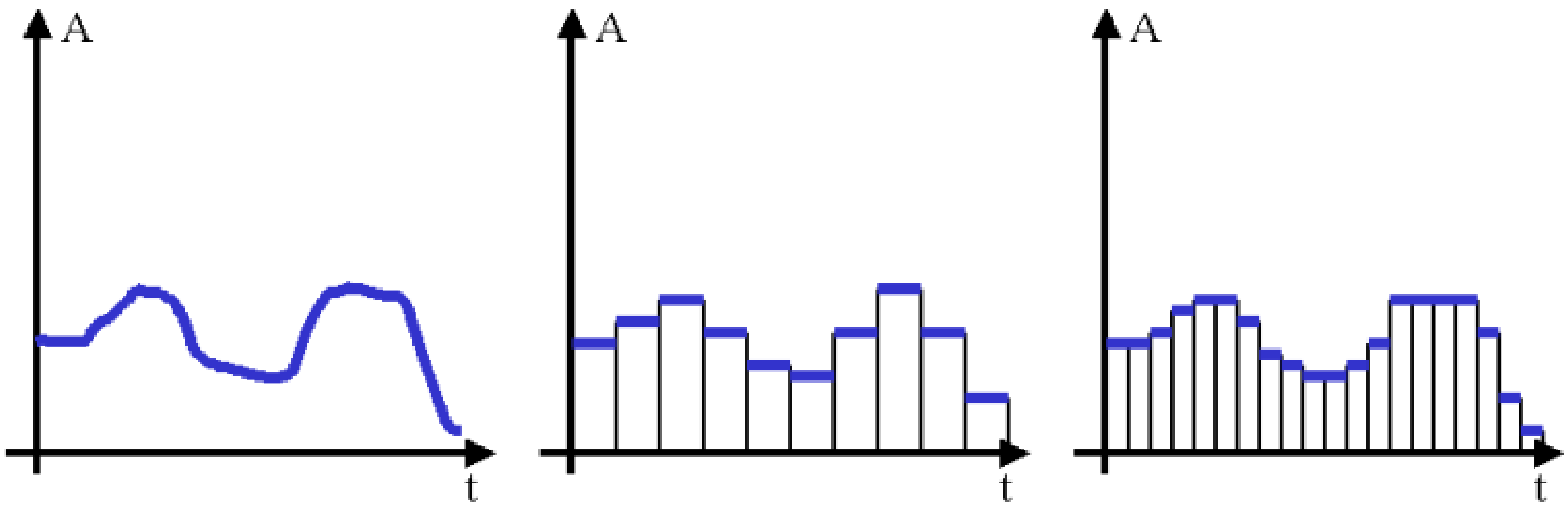
Organization of Lecture

1. Digital and Analog Signal.
2. Analog to Digital and Digital to Analog Conversion.
3. Analog to Digital Converter.
4. Digital to Analog Converter.
5. Aliasing and Sampling.
6. Quantization and Coding.

1. Digital and Analog Signal

Digital Signals are discrete-time signals having a set of discrete values.

Analog signals are defined for every value of time and they take on values in the continuous interval.

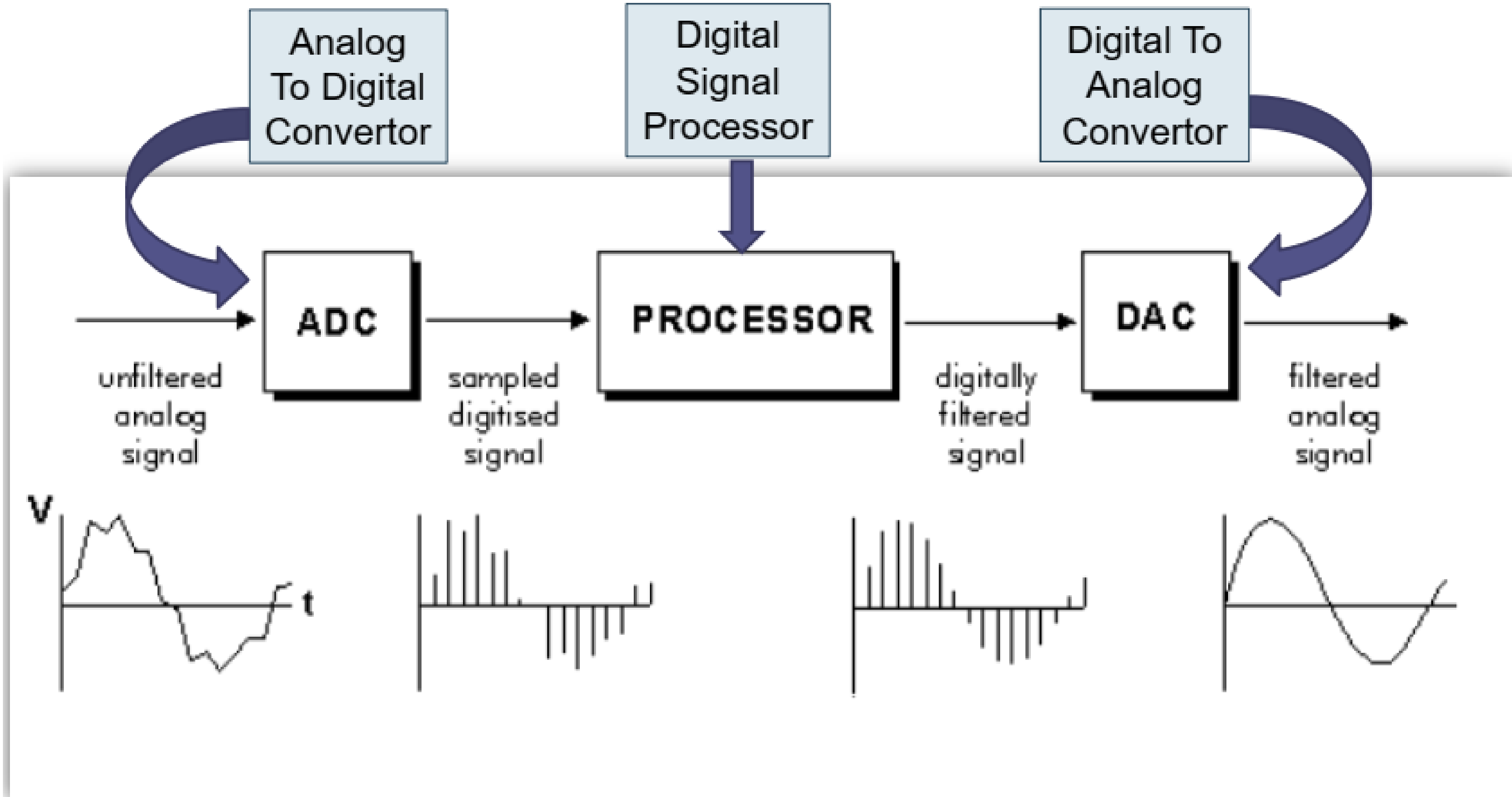


Analog signal – continuously varying

Digital signal – large time divisions

Digital signal – small time divisions

2. Analog to Digital and Digital to Analog Conversion

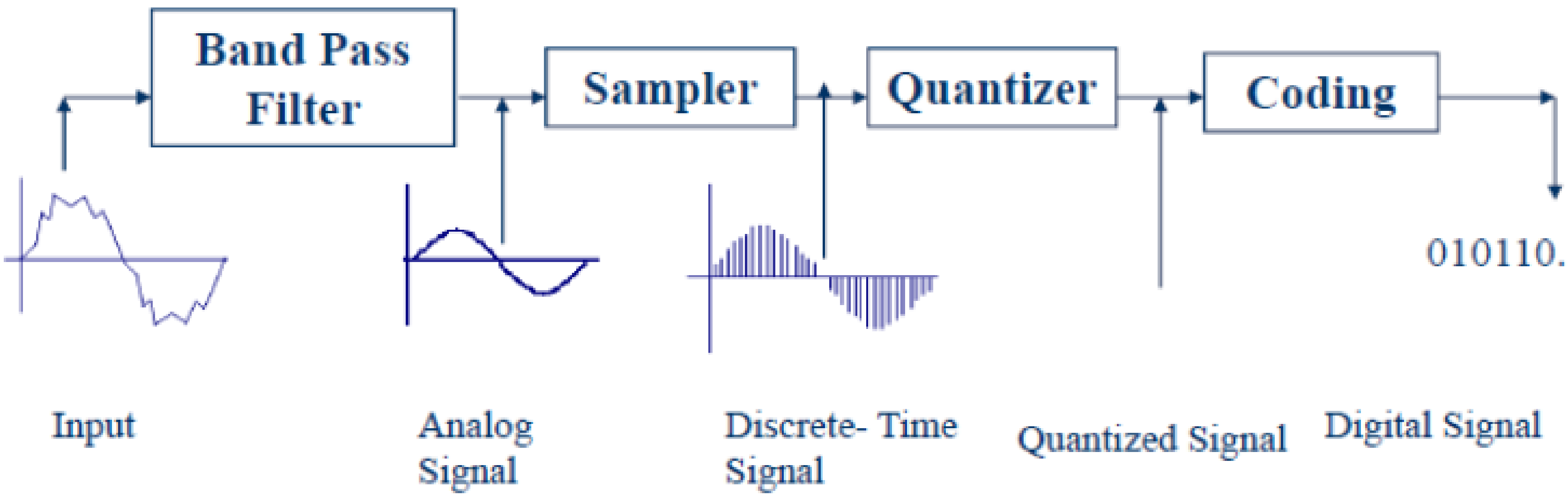


3. Analog to Digital Converter

An analog-to-digital converter (ADC) is a device that converts an analog voltage into a digital number.

Steps Of A/D Conversion:

1. Sampling.
2. Quantization.
3. Coding.

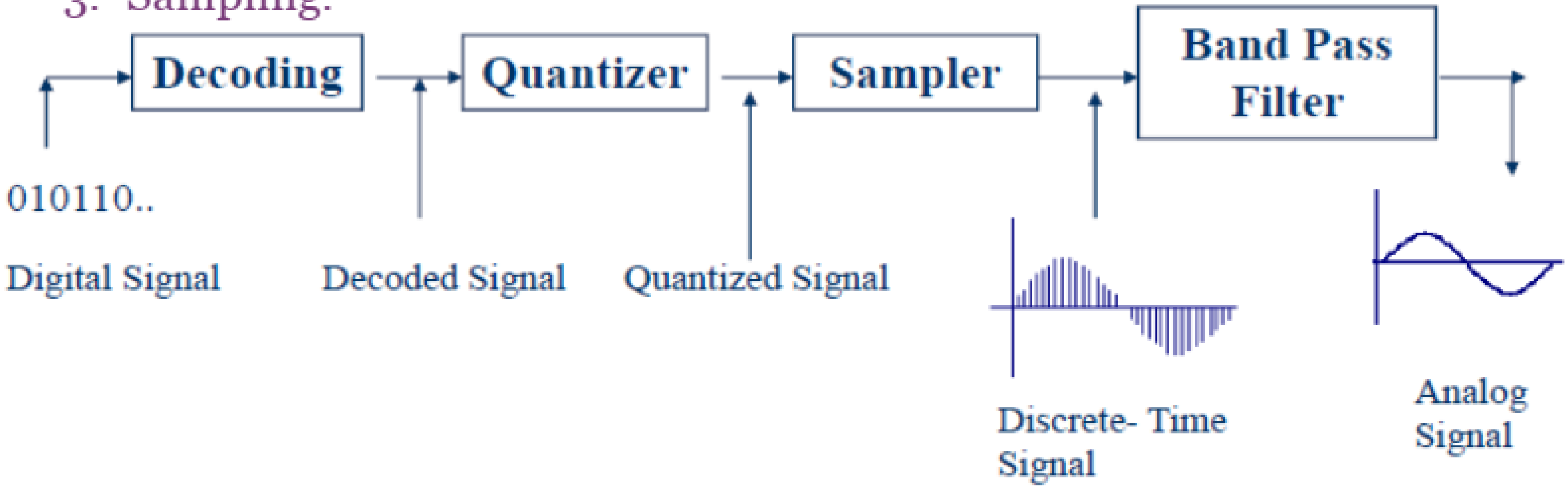


4. Digital to Analog Converter

A digital-to-analog converter(DAC) converts a digital number to an analog voltage.

Steps of D/A Conversion:

1. Decoding.
2. Quantization.
3. Sampling.



5. Aliasing and Sampling

- The ***sampling*** is the process of conversion of a continuous time signal into a discrete time signal. The sampling is performed by taking samples of continuous time signal at definite intervals of time. Usually, the time interval between two successive samples will be same and such type of sampling is called ***periodic or uniform sampling***.
- The time interval between successive samples is called ***sampling time*** (or sampling period or sampling interval), and it is denoted by "T". The unit of sampling period is second (s).
- The inverse of sampling period is called ***sampling frequency*** (or sampling rate), and it is denoted by F_s. The unit of sampling frequency is Hertz (Hz).

5. Aliasing and Sampling

Conversion of Continuous Time Signal to Discrete Time Signal:

Continous Time Signal.

$x_a(t) = x(t) = A\cos(\Omega t + \phi)$

where:

Ω = Analog frequency = $2\pi F$ (in rad/sec)

$x(t) = A\cos(2\pi F t + \phi)$ — (i)

Discrete Time Signal

Put $t = nT$ in eq/(i) we get a D.T signal

$x(nT) = A\cos(2\pi F(nT) + \phi)$ — (ii)

Compare eq (i) & (ii)

$x(t) = x(nT)$

$A\cos(2\pi F t + \phi) = A\cos(2\pi F(nT) + \phi)$

$t = nT$

$\therefore T = \frac{1}{F_s}$; where F_s = Sampling Frequency

5. Aliasing and Sampling

so $t = n \times \frac{1}{F_s}$

put $t = \frac{n}{F_s}$ in eq -(i)

$$x\left(\frac{n}{F_s}\right) = A\cos\left(2\pi F \times \left(\frac{n}{F_s}\right) + \phi\right)$$

$$x\left(\frac{n}{F_s}\right) = A\cos\left(2\pi\left(\frac{F}{F_s}\right)n + \phi\right) \text{ --- (iii)}$$

5. Aliasing and Sampling

General Equation of D.T Signal

$$x(n) = A\cos(2\pi f n + \phi)$$

Compare it with eq (iii) we get:

$$f = \frac{F}{F_s}$$

where f = normalized frequency is a frequency in units of cycles/sample or radians/sample.

Recall

$$\omega = 2\pi f$$

ω = Digital Frequency

$$\omega = 2\pi\left(\frac{F}{F_s}\right)$$

$$\omega = 2\pi F \times \frac{1}{F_s}$$

$$\Omega = 2\pi F$$

$$T = \frac{1}{F_s}$$

$$\omega = \Omega T$$

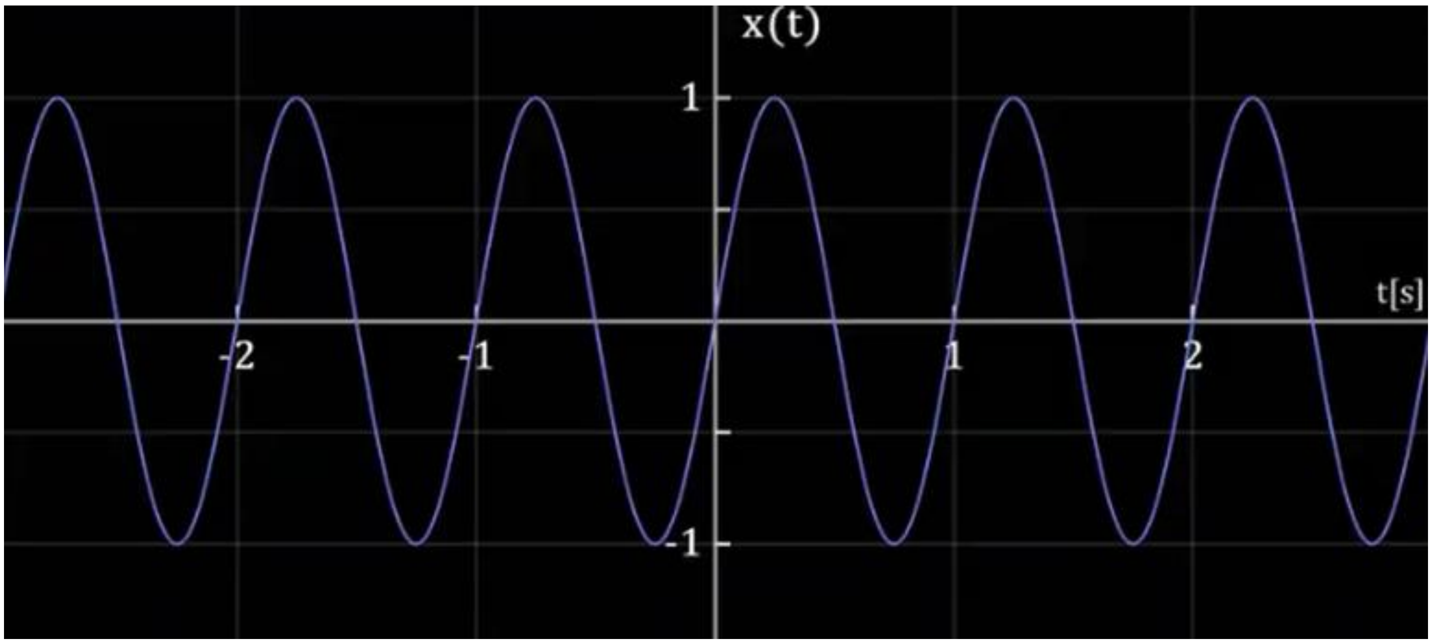
Also

F = Frequency of original C.T signal

F_s = Sampling frequency.

5. Aliasing and Sampling

* Before going into our main topic; let's just review the notion of frequency for C.T signal:

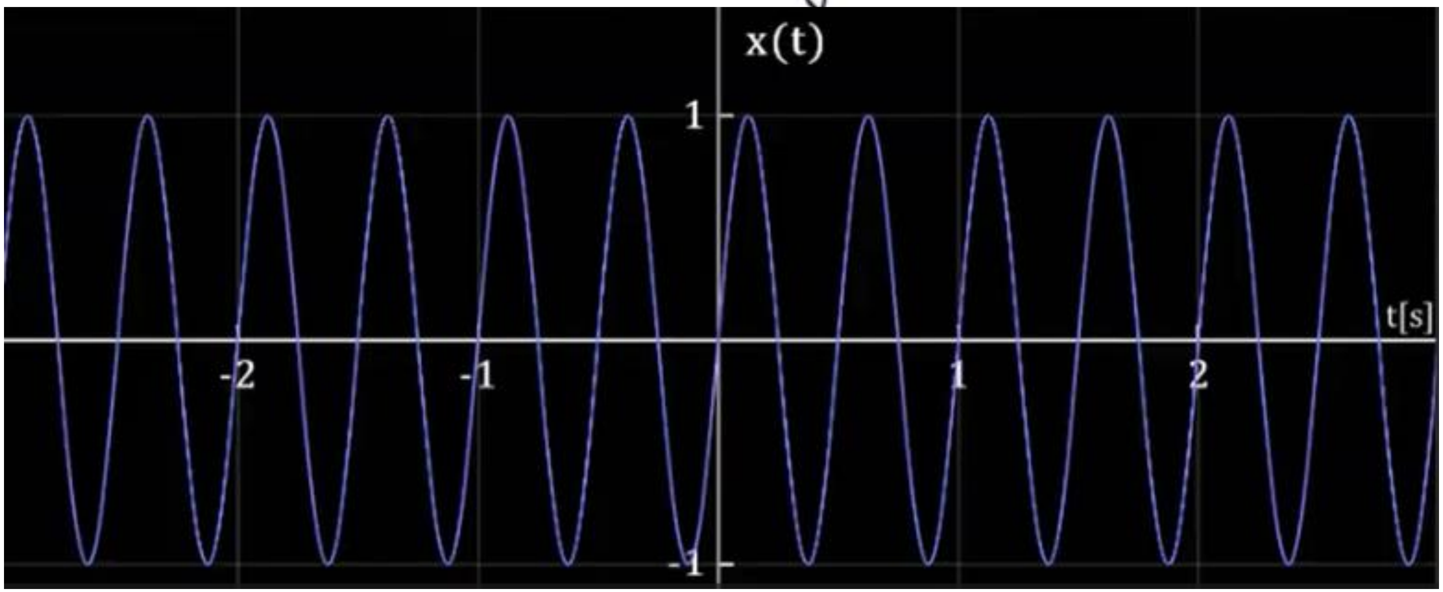


* The above signal completes 1 cycle in 1 sec, so the frequency of C.T signal is:

$$F = 1 \text{ cycle/s} = 1 \text{ Hz.}$$

5. Aliasing and Sampling

* Now consider another signal:

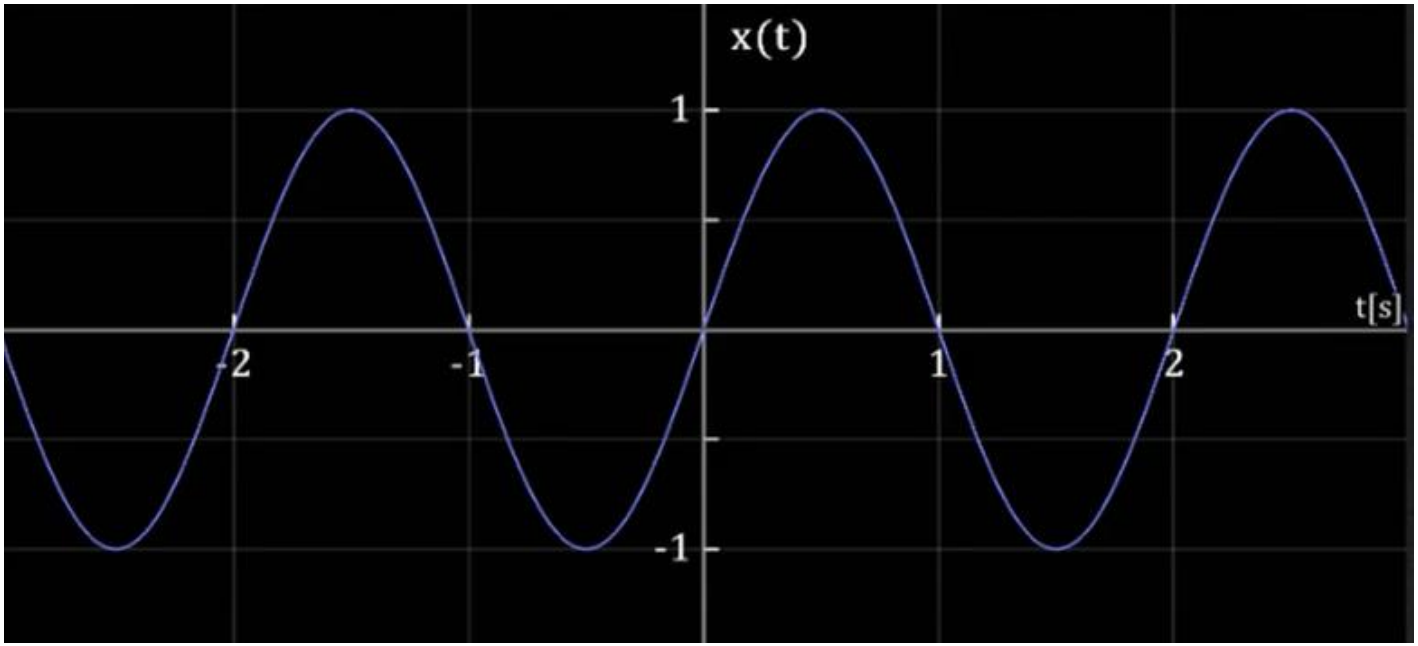


Here the signal completes 2 cycles in 1sec, so the frequency of C.T signal is:

$$F = 2\text{cycles/sec} = 2\text{ Hz}$$

5. Aliasing and Sampling

* Now consider another signal :

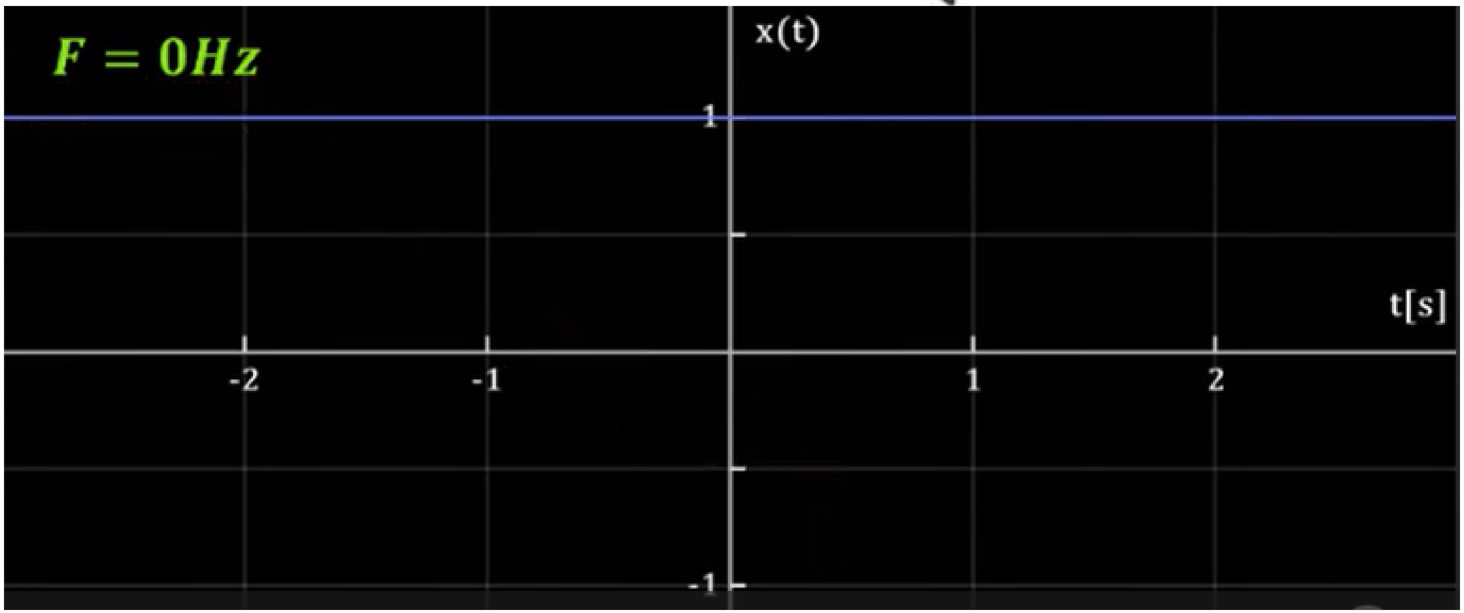


If the signal completes Half cycle in 1 sec, the frequency of C.T signal is

$$F = 0.5 \text{ cycles/sec} = 0.5 \text{ Hz} = \frac{1}{2} \text{ Hz}.$$

5. Aliasing and Sampling

* Now consider another signal :



If the signal is not varying repeatedly with respect to time, the frequency of C.T will be :

$F = 0Hz.$

5. Aliasing and Sampling

* Range of frequency for continuous Time signals is

$$-\infty < F < \infty$$

* But this is not the case with Discrete Time signals *

* Recall!

Normalised Frequency, $f = \frac{F}{F_s}$

where;

F = Frequency of C.T signal

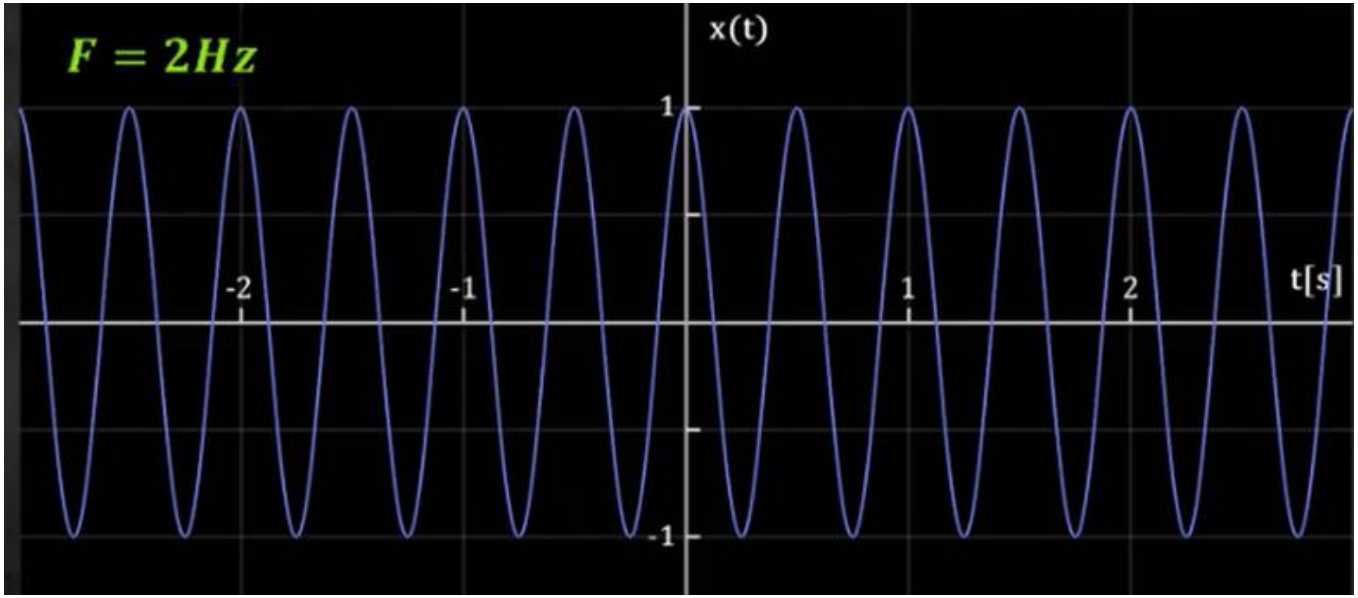
F_s = Sampling Frequency / Sampling Rate

5. Aliasing and Sampling

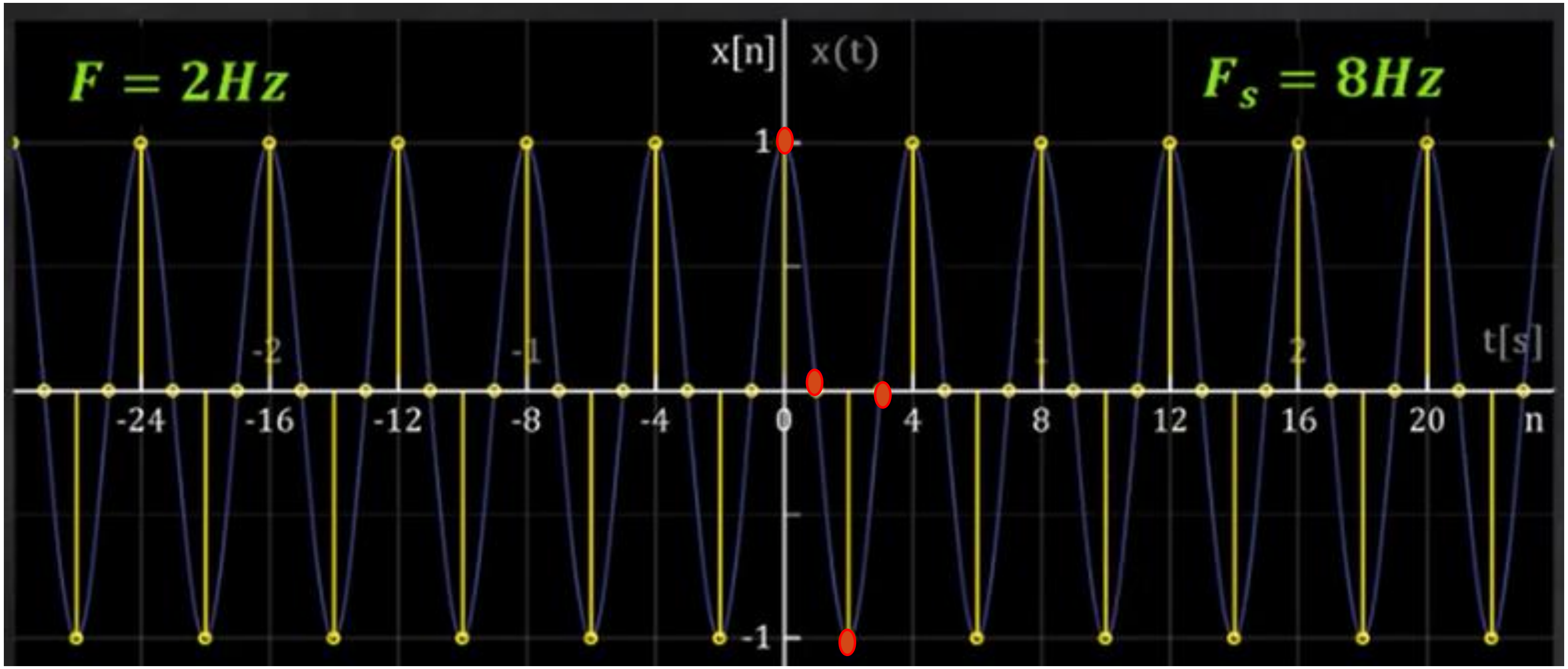
* Now let's proceed further, Given below is the C.T cosine signal with a frequency of 2Hz ($F=2Hz$) and sampling rate of 8 Hz ($F_s=8Hz$). Therefore resulting normalized frequency will be:

$$f = \frac{F}{F_s} = \frac{2Hz}{8Hz} = \frac{1}{4}$$

; which means it takes 4 samples to complete 1 cycle of D.T signal. OR we get 4 samples per cycle of D.T signal.



5. Aliasing and Sampling



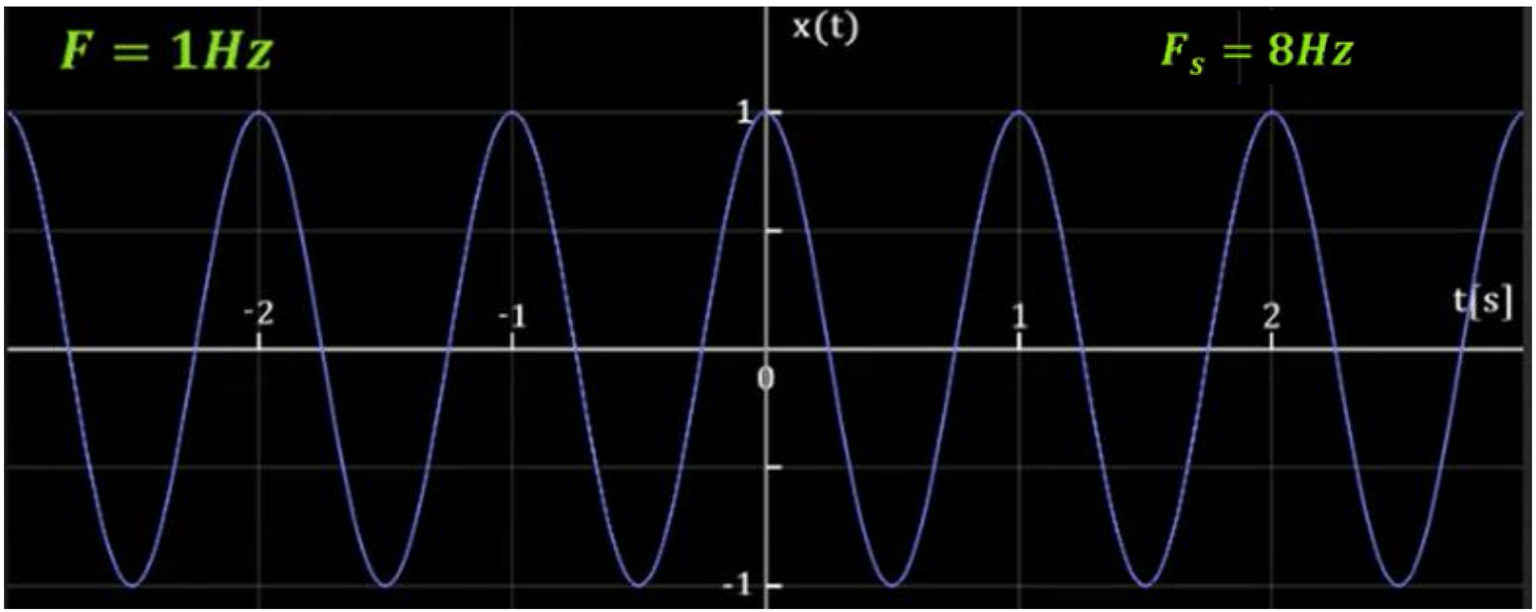
$$f = \frac{F}{F_s} = \frac{2}{8} = \frac{1}{4}$$

Here, with F = 2Hz and F_s= 8Hz. We get 4 samples per cycle of D.T signal.

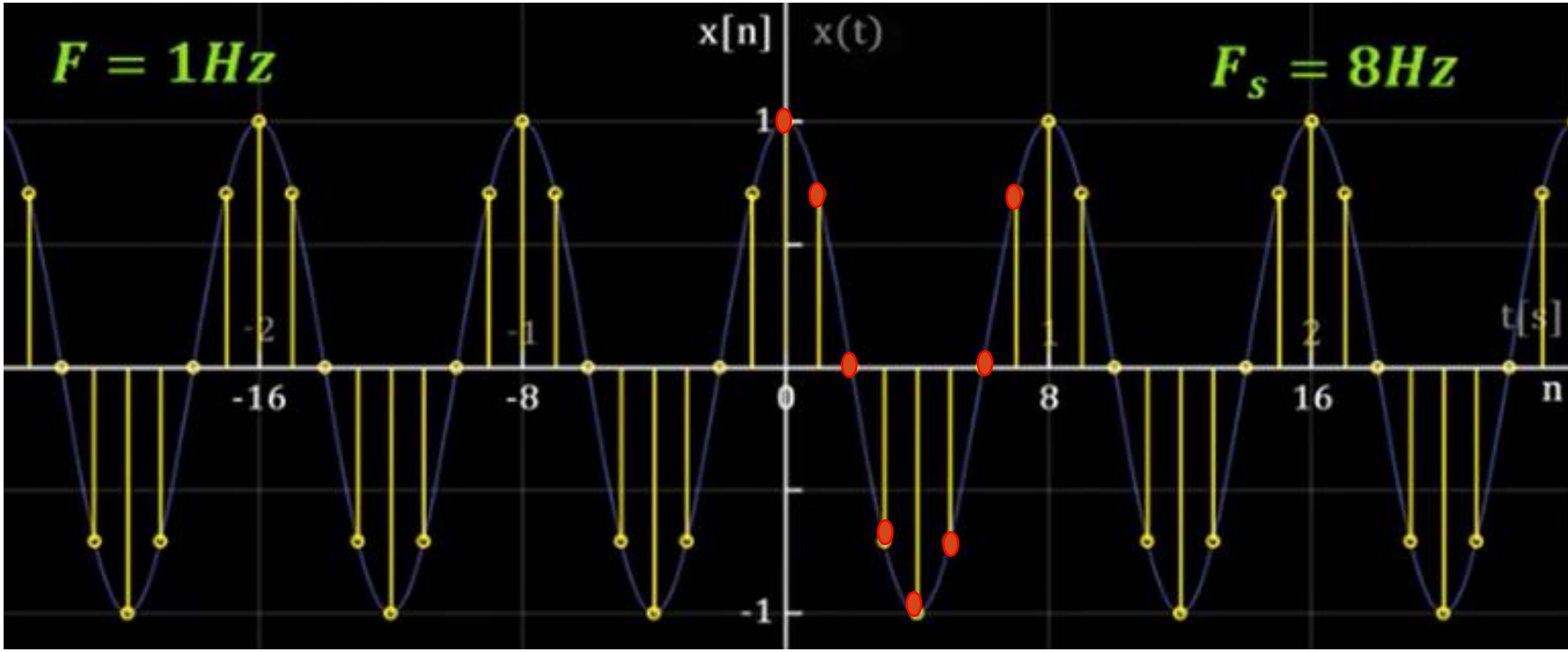
5. Aliasing and Sampling

* Now consider another C.T cosine signal with a frequency of 1 Hz and sampling frequency of 8 Hz & resulting normalized frequency is:

$$f = \frac{1}{8Hz} = \frac{1}{8}$$; it takes 8 samples to complete 1 cycle of D.T signal.



5. Aliasing and Sampling

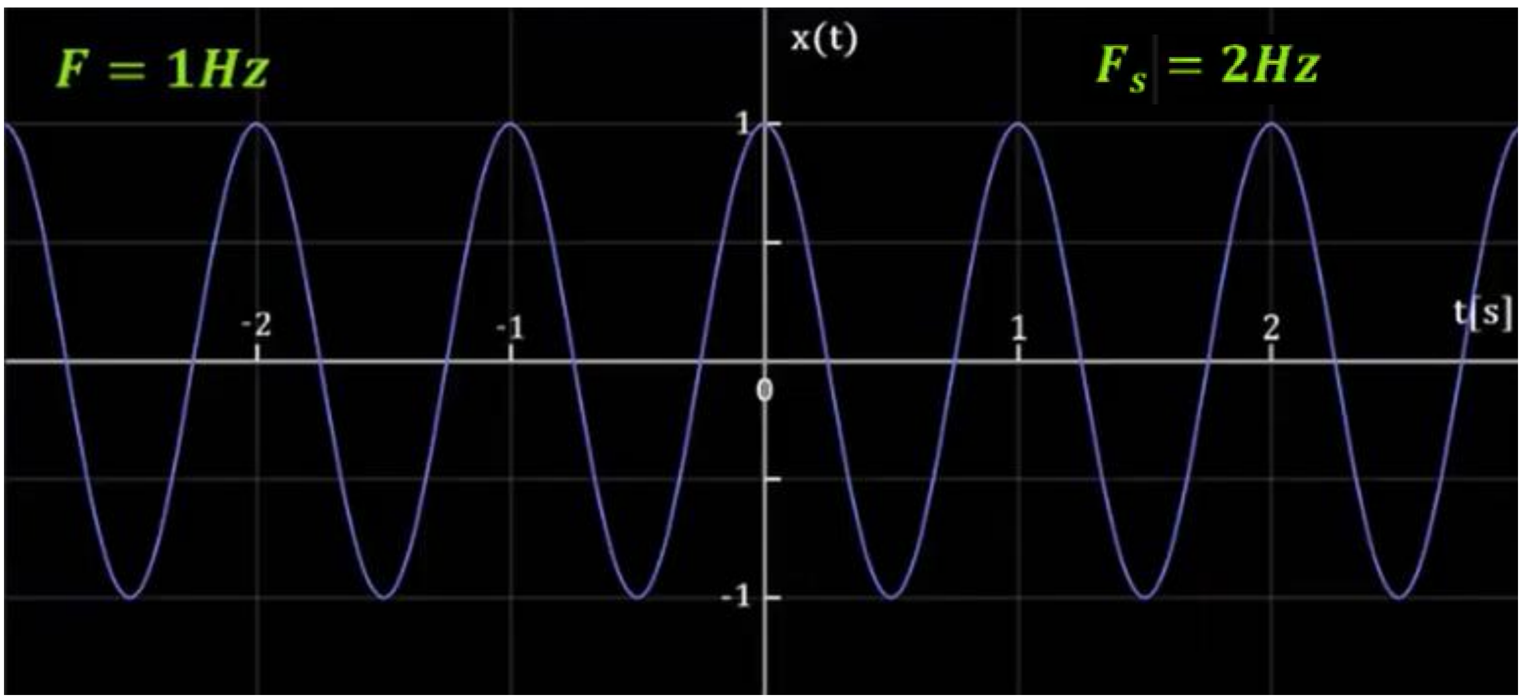


Here, with F = 1Hz and F_s= 8Hz. We get 8 samples per cycle of D.T signal.

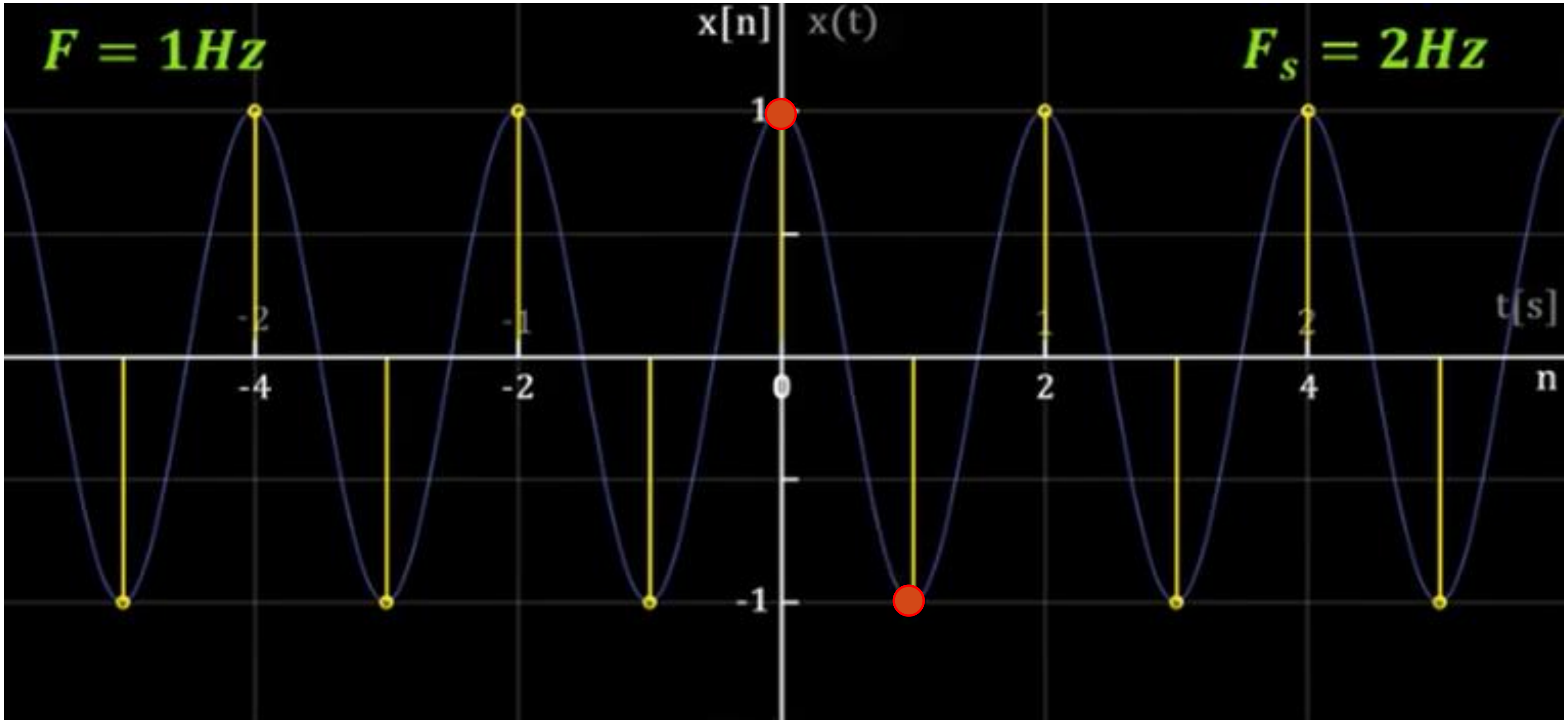
5. Aliasing and Sampling

* Now let us trying sampling the same signal with a sampling frequency of 2 Hz i.e, $F = 1\,Hz$ & $F_s = 2\,Hz$ and the resulting normalized is:

$f = \frac{1}{2}$; it takes 2 samples to complete 1 cycle of D.T signal.



5. Aliasing and Sampling

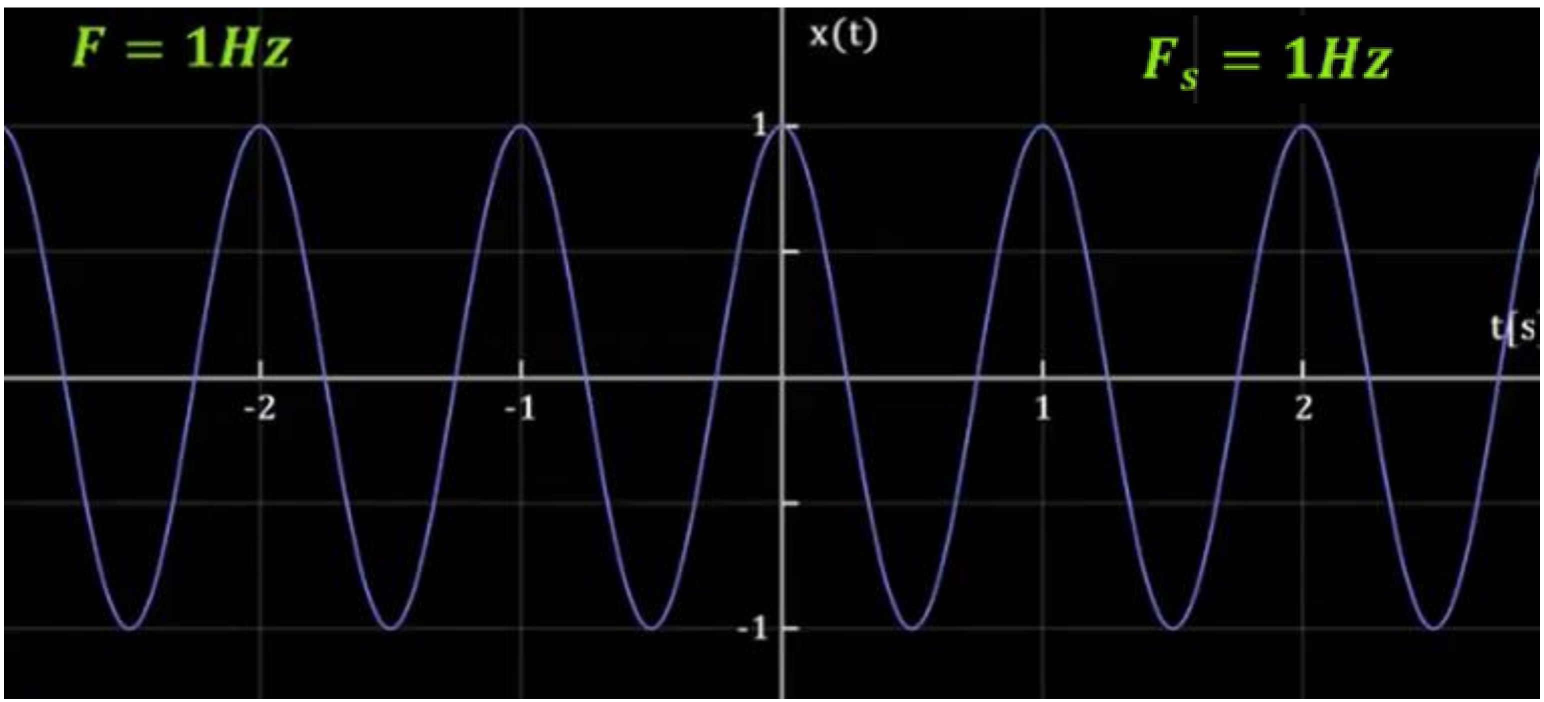


Here, with F = 1Hz and F_s= 2Hz. We get 2 samples per cycle of D.T signal.

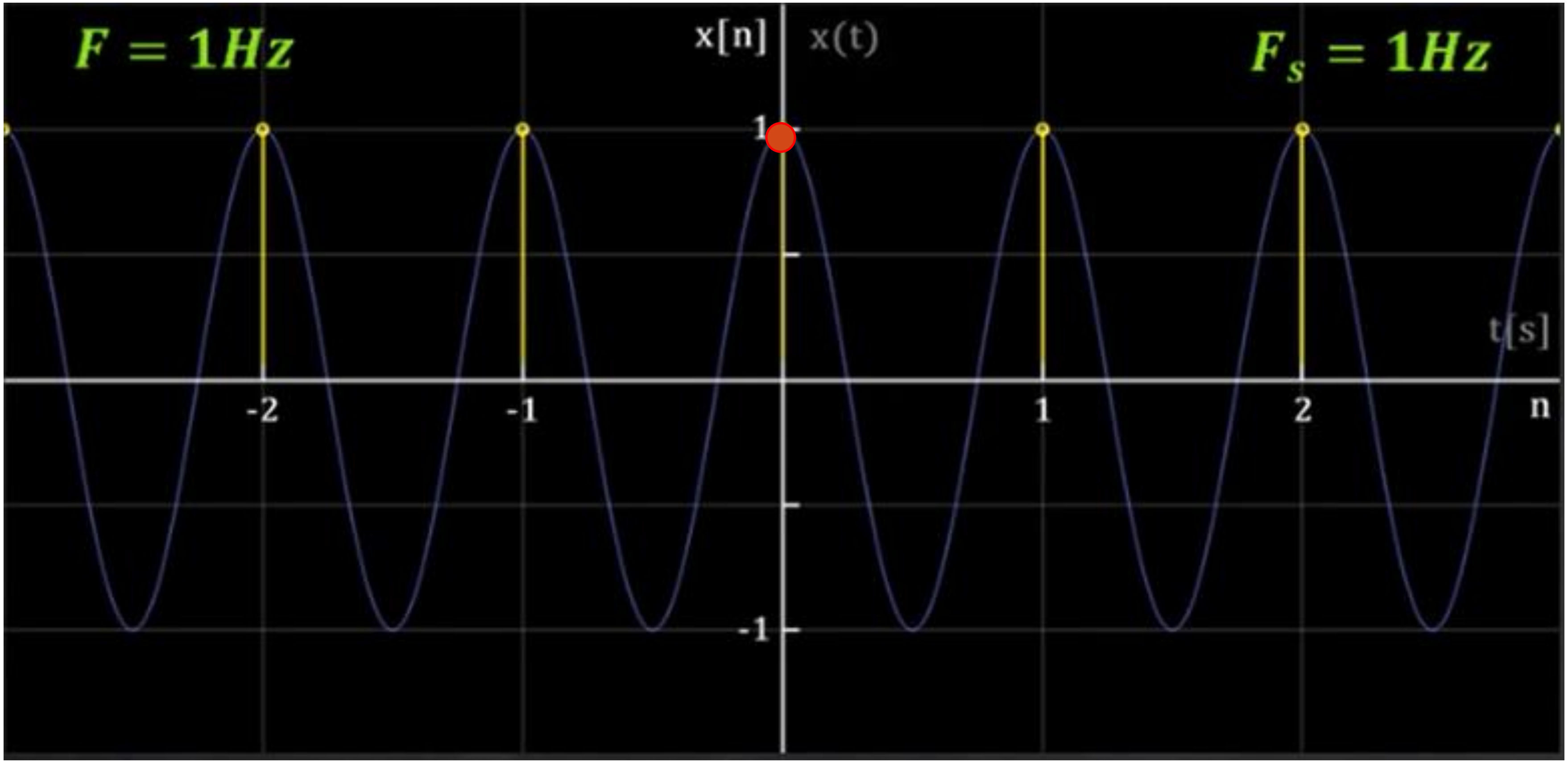
5. Aliasing and Sampling

* Again if the same C.T cosine signal with frequency of 1 Hz and sampling frequency of 1 Hz. The normalized frequency will be:

$f = \frac{1}{1} = 1\,Hz$; it takes 1 sample to complete 1 cycle of D.T signal.



5. Aliasing and Sampling



Here, with F = 1Hz and F_s= 1Hz. We get 1 sample per cycle of D.T signal.

5. Aliasing and Sampling

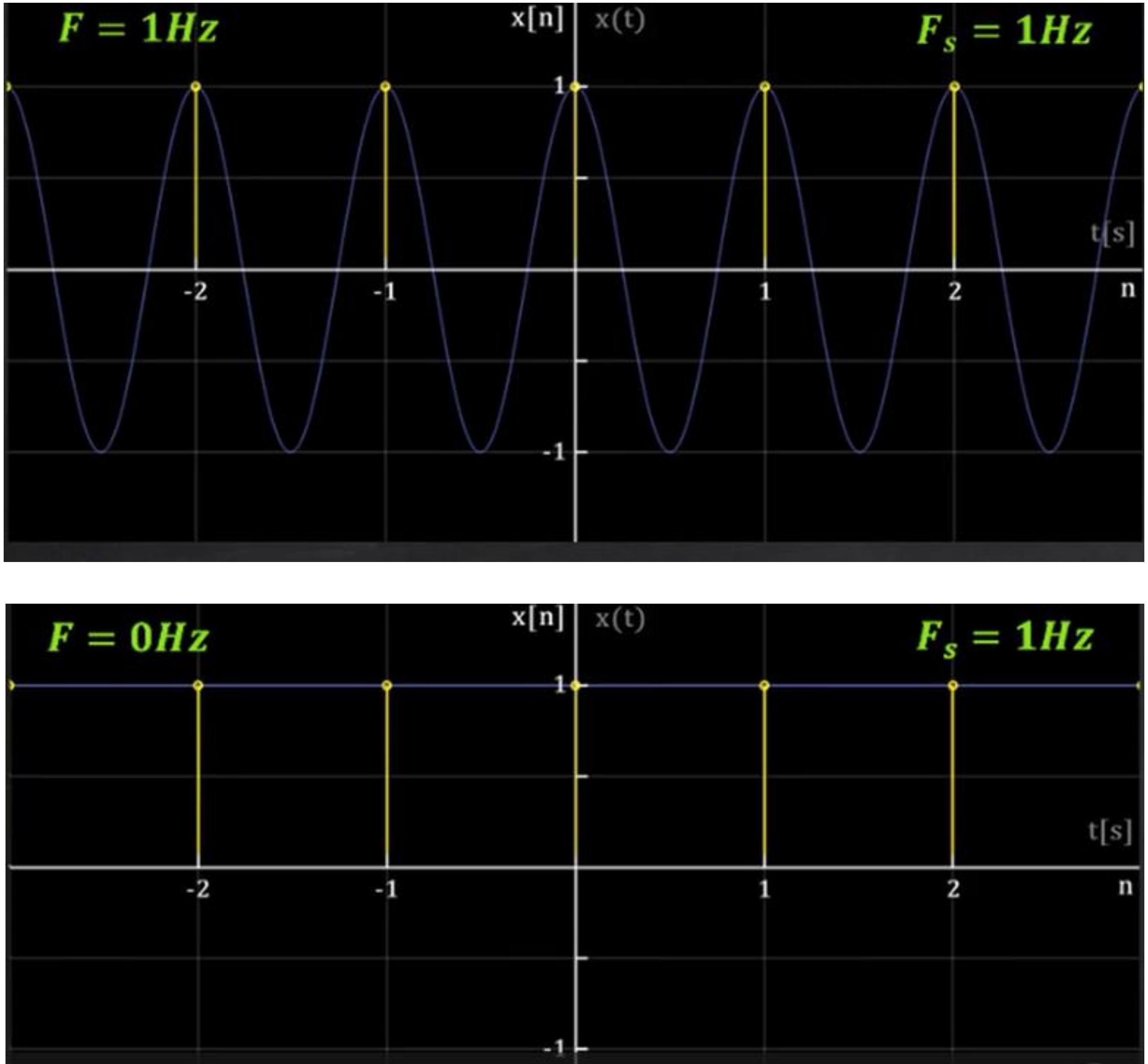
* Now consider the continuous time signal with a frequency of zero hertz and sampling frequency 1 Hz. The resulting normalized frequency is:

$$f = \frac{0}{1} = 0;$$

* But the two discrete type signals i.e., $f = \frac{1}{1} = 1\,Hz$ and $f = \frac{0}{1} = 0$ are the same.

Two plots are shown in next slides

5. Aliasing and Sampling



The two D.T signals are same

5. Aliasing and Sampling

* What we can do about it??

> We can choose to say that a D.T signal cannot have frequencies larger than $\frac{1}{2}$. So the range of frequencies possible for D.T signals are:

$$\boxed{-\frac{1}{2} \leq f \leq \frac{1}{2}}$$

* The highest rate of oscillation in a D.T sinusoid is obtained when:

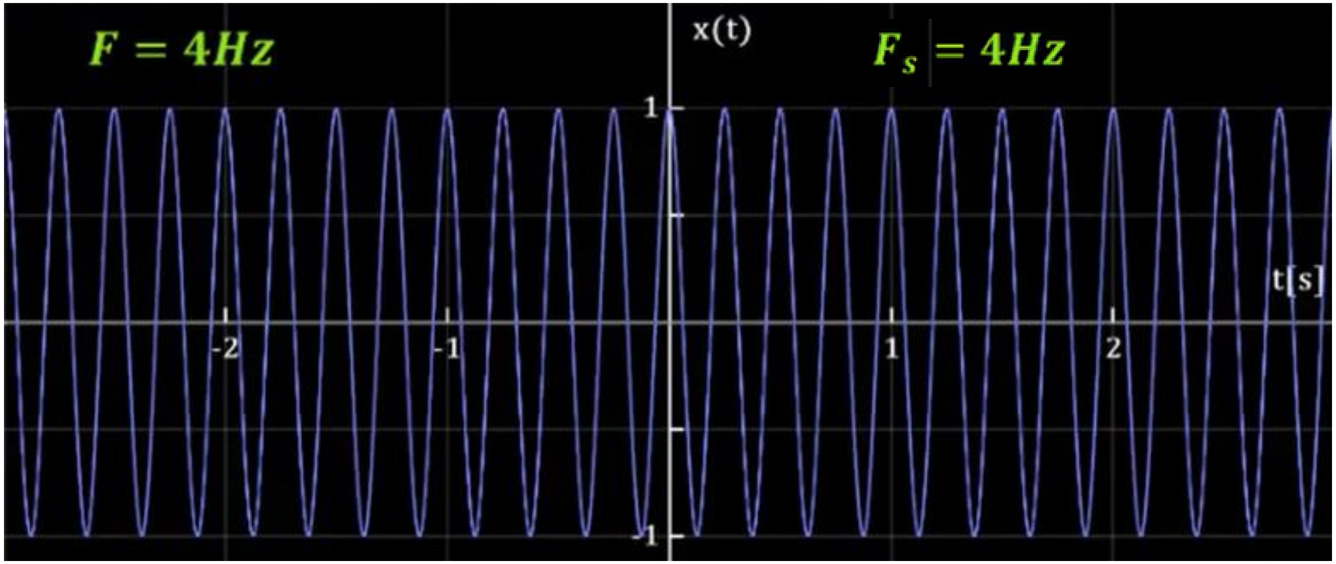
$$f = \frac{1}{2} \left(\text{or } f = -\frac{1}{2}\right).$$

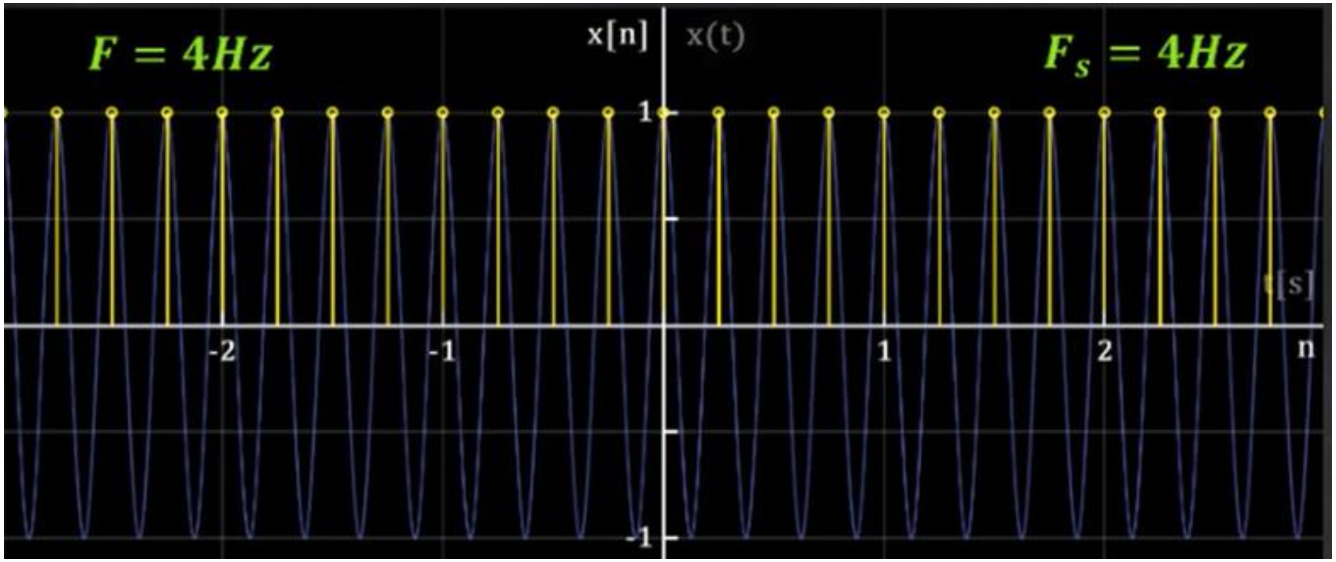
5. Aliasing and Sampling

* Another interesting phenomena of a D.T signal.
Consider a C.T sinusoid signal with frequency of 4 Hertz and Sampling frequency of 4 Hz, the normalized frequency will be?

$$f = \frac{4\,Hz}{4\,Hz} = 1\,Hz.$$

5. Aliasing and Sampling



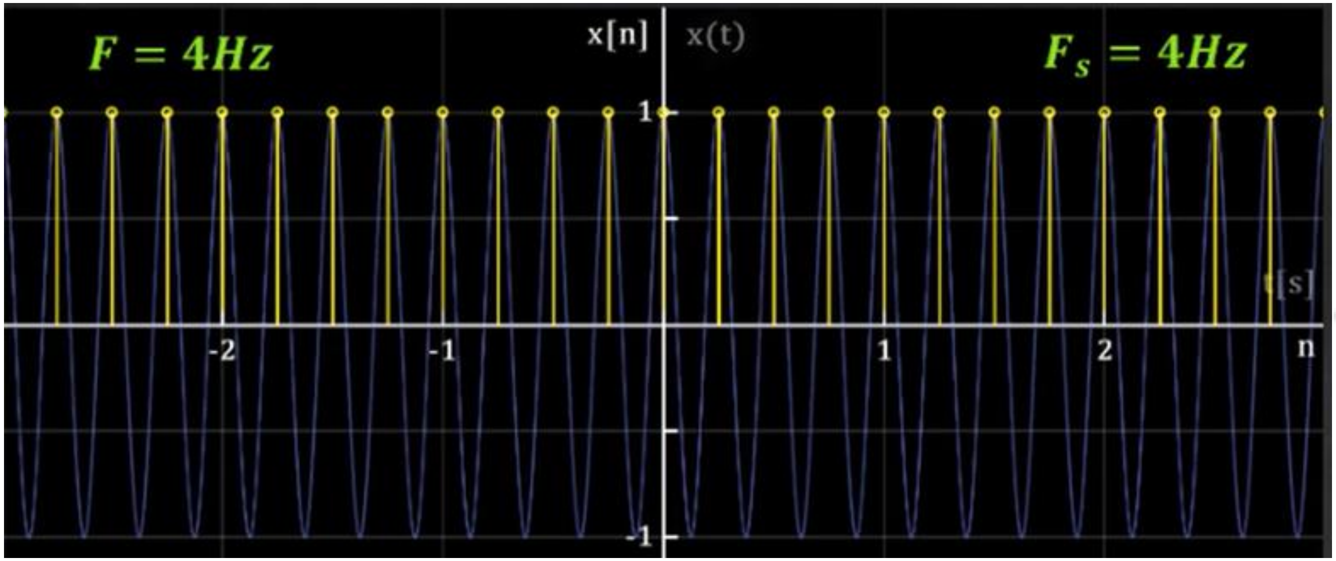


5. Aliasing and Sampling

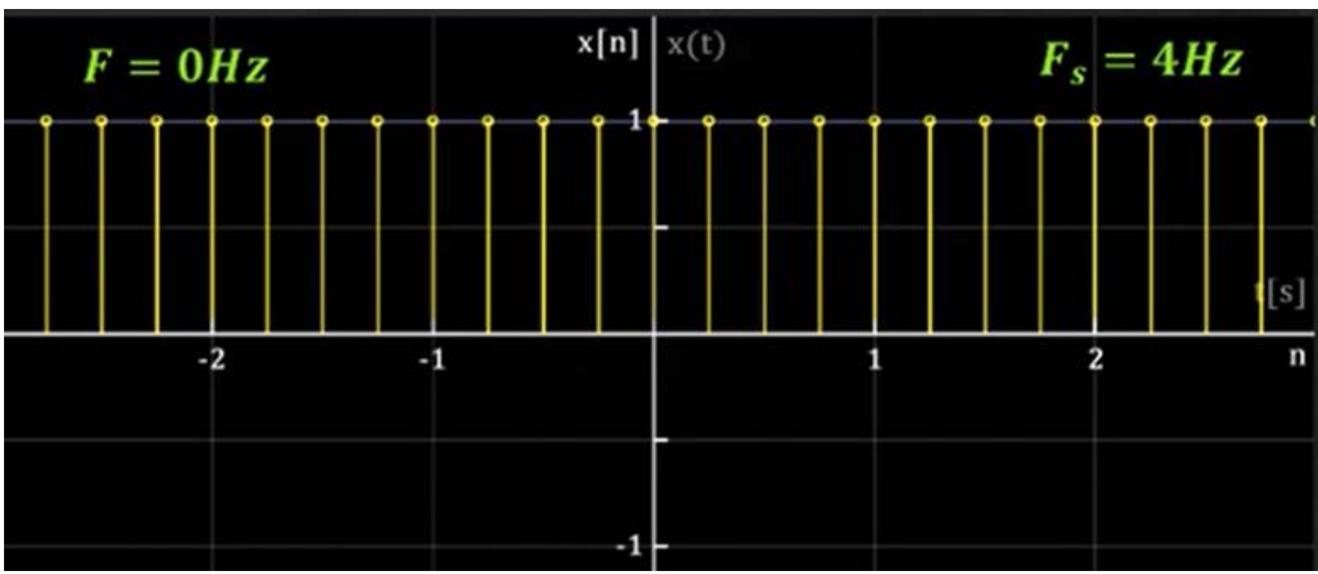
* Let's say we have another signal with frequency of 0 hertz, the normalized frequency will be:

$$f = \frac{0}{4} = 0.$$

5. Aliasing and Sampling



$$f = \frac{F}{F_s} = \frac{4}{4} = 1$$



$$f = \frac{F}{F_s} = \frac{0}{4} = 0$$

5. Aliasing and Sampling

* Any two normalized frequencies that differ by an integer will be identical or will result in same D.T signal i.e.,

$$\boxed{f = f_0 + k, \quad k \rightarrow \text{integer}}$$

* let's take an example to validate this concept
> Consider a C.T cosine signal with $F = 1\,Hz$ & $F_s = 2\,Hz$. The normalized frequency will be

$$f_0 = \frac{1}{2}$$

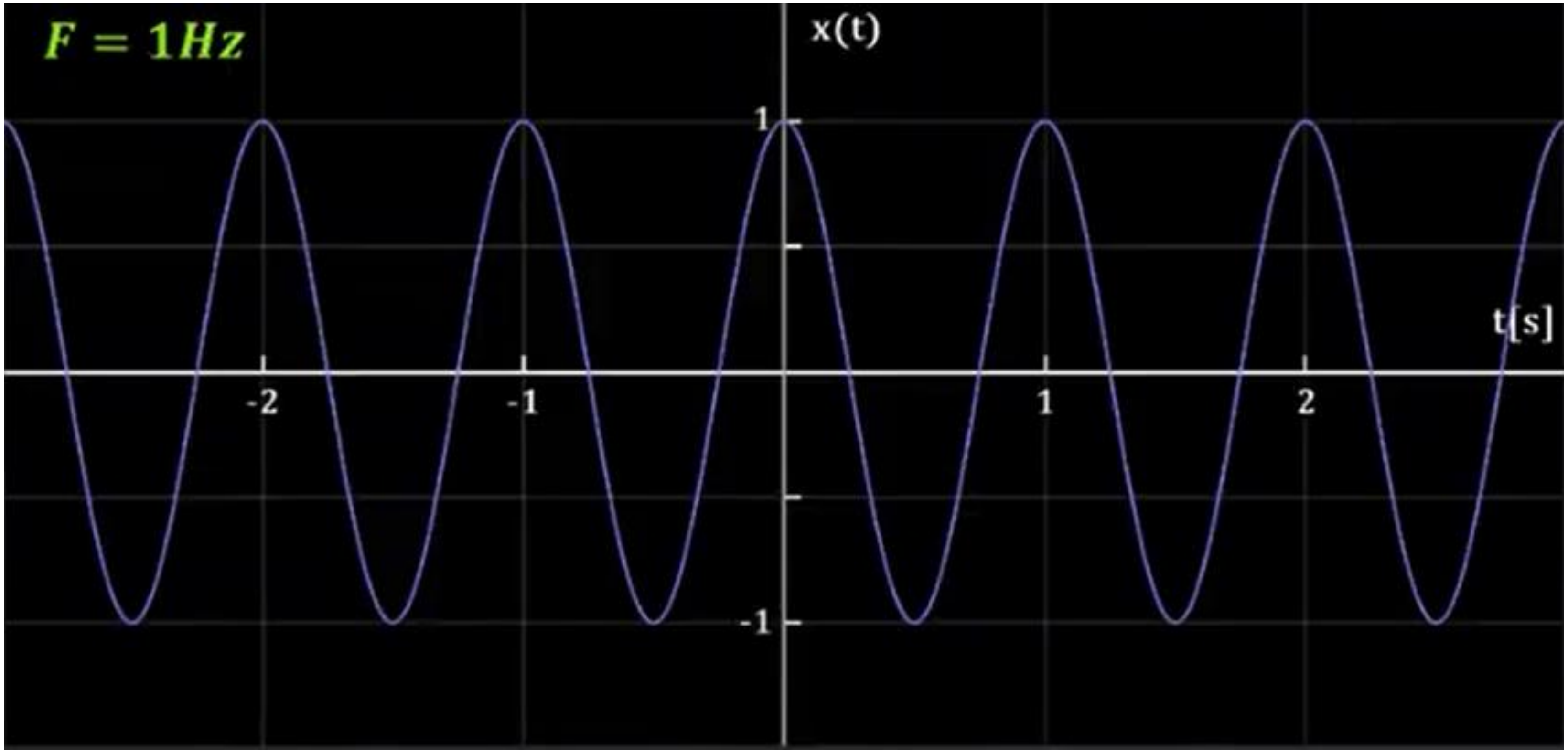
> Now if we add 1 to this normalized frequency i.e.,

$$f = \frac{1}{2} + 1 = \frac{3}{2}$$

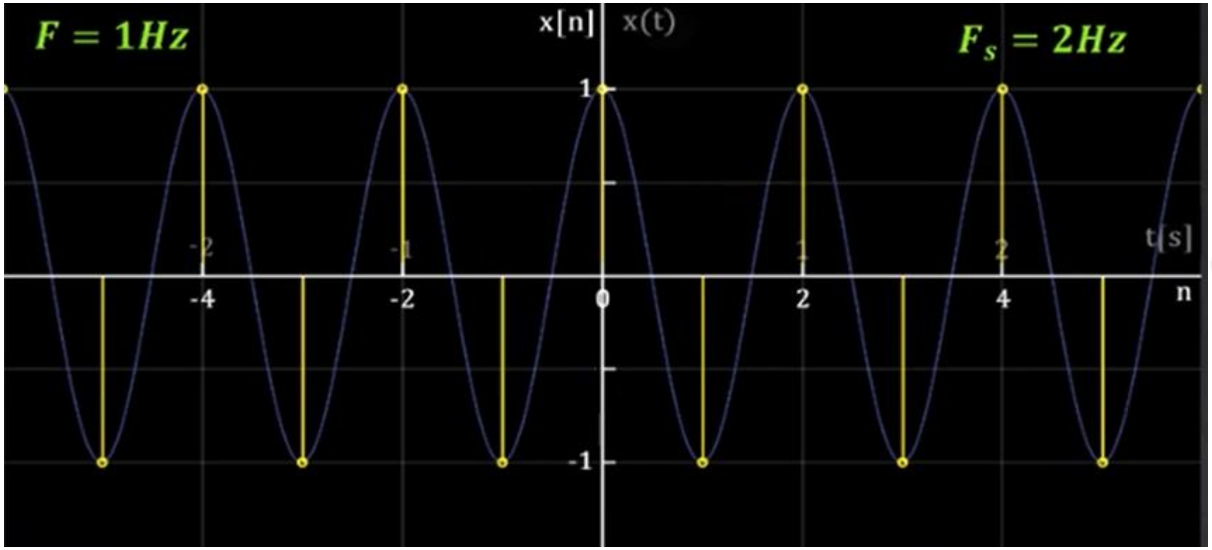
> f_0 & f will be identical as per $f = f_0 + k$

> This phenomenon is called Aliasing

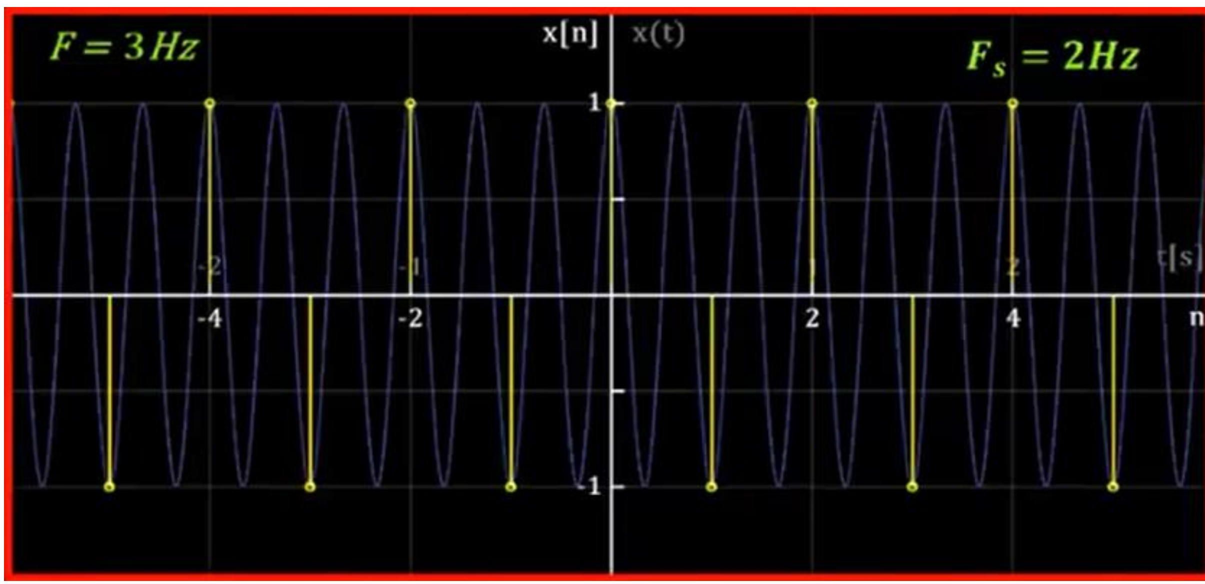
5. Aliasing and Sampling



5. Aliasing and Sampling



$f_0 = 1/2$



$f = f_0+1 = \frac{1}{2}+1=3/2$

5. Aliasing and Sampling

Aliasing

An effect that causes diffuent signals to become indistinguisable (identical) when sampled

Sampling Theorem

It states that sampling frequency (F_s) should be greater than or equal to twice the highest frequency (F_{max}) component of the input signal

$$\boxed{F_s \geq 2F_{max}}$$ ⇒ To prevent Aliasing

* Any sampling frequency (F_s) less than twice the input signal frequency will cause aliasing effect

5. Aliasing and Sampling

<u>Nyquist Rate</u>

When sampling frequency equals twice the input signal frequency, that sampling rate is called Nyquist Rate.

* The relationship between analog and digital frequency is:

$$f = \frac{F}{F_s}$$

> The range of discrete frequency to avoid aliasing

$$-\frac{1}{2} \leq f \leq \frac{1}{2}$$

$$-\frac{1}{2} \leq \frac{F}{F_s} \leq \frac{1}{2} \qquad \because f = \frac{F}{F_s}$$

> On multiplying above equation with F_s we get.

$$-\frac{F_s}{2} \leq F \leq \frac{F_s}{2}$$

5. Aliasing and Sampling

Example 1:

Given below are the Continuous Time (C.T) signals i.e., $x_1(t)$ & $x_2(t)$. Convert them into discrete time (D.T) signals i.e., $x_1(n)$ & $x_2(n)$. Both C.T signals are sampled at 40 Hz.

$$x_1(t) = \cos 2\pi(10)t$$

&

$$x_2(t) = \cos 2\pi(50)t$$

Solution:

Given

F_s = Sampling frequency/sampling rate = 40 Hz

Consider signal $x_1(t)$:

$$x_1(t) = \cos 2\pi(10)t \quad \text{— (i)}$$

Recall!

$$x(t) = A\cos(\Omega t + \phi)$$

$$\therefore \quad \Omega = 2\pi F$$

$$x_1(t) = A\cos(2\pi F_1 t + \phi) \quad \text{— (ii)}$$

5. Aliasing and Sampling

compare (i) & (ii)

$$x_1(t) = A\cos(2\pi F_1 t + \phi)$$

$$x_1(t) = \cos(2\pi 10 t + \phi$$

we get: $F = F_1 = 10\,Hz$

Also Recall!

$$x_1(n) = A\cos(2\pi f_1 n + \phi) \text{ --- (iii)}$$

where:

$$f_1 = \frac{F_1}{F_s} = \frac{10\,Hz}{40\,Hz} = \frac{1}{4}\,Hz = 0.25\,Hz$$

normalized frequency "f_1" is bounded b/w:

$$= -\frac{1}{2} \le f_1 \le \frac{1}{2} = -\frac{1}{2} \le f_1 \le \frac{1}{2} = -0.5 \le 0.25 \le 0.5$$

eq (iii)

$$\boxed{x_1(n) = A\cos\left(2\pi\left(\frac{1}{4_2}\right)n\right) = \cos\left(\frac{\pi}{2}\right)n}$$

5. Aliasing and Sampling

Consider signal $x_2(t)$:

$$x_2(t) = \cos 2\pi(50)t$$
$$x_2(t) = A\cos(2\pi F_2 t + \phi)$$

on comparison we get:

$$F_2 = 50\,Hz$$

Also

$$x_2(n) = A\cos(2\pi f_2 n + \phi) \quad \text{—(iv)}$$

where:

$$f_2 = \frac{F_2}{F_s} = \frac{50}{40} = \frac{5}{4} = 1.25$$

normalized frequency f_2 is not bounded b/w limit i.e., b/w $-0.5 \le f_2 \le 0.5$. We need to normalize f_1 such that it satisfies the condition. Therefore.

$$f_2 = \frac{5}{4} \quad \text{-(v)}$$

$$f_2 = 1 \pm \frac{x}{4} \quad \text{—(vi)}$$

5. Aliasing and Sampling

$$\frac{5}{4} = 1 \pm \frac{x}{4}$$

$$\frac{5}{4} - 1 = \pm \frac{x}{4}$$

$$\frac{1}{4} = \pm \frac{x}{4}$$

$$\cancel{4} \times \frac{1}{\cancel{4}} = \pm x$$

So $x = \pm 1$

put $x = +1$ in eq -(vi)

$f_2 = 1 + \frac{1}{4} = \frac{5}{4} \rightarrow$ which is same as eq-(v)

put $x = -1$ in eq -(vi)

$f_2 = 1 - \frac{1}{4} = \frac{3}{4} \rightarrow$ which is not same as eq-(v)

Consider case when $\underline{x = +1}$

5. Aliasing and Sampling

eq (iv)

$$x_2(n) = \cos\left(2\pi\left(1+\frac{1}{4}\right)n\right)$$

$$= \cos\left(2\pi + \frac{2\pi}{4}\right)n$$

$$= \cos\left(\frac{2\pi}{4_2}\right)n$$

$$\boxed{x_2(n) = \cos\left(\frac{\pi}{2}\right)n}$$

We know that D.T sinusoids whose frequencies are separated by integer multiple of 2π are identical.

5. Aliasing and Sampling

Example 2:

Consider the analog signal:

$$x_a(t) = 3\cos(100\pi t)$$

(a) Determine the minimum sampling rate required to avoid aliasing.

(b) Suppose that the signal is sampled at the rate $F_s = 200$ Hz. What is the discrete-time signal obtained after sampling?

(c) Suppose the signal is sampled at the rate $F_s = 75$ Hz. What is discrete-time signal obtained after sampling?

(d) What is the frequency $0 < F < F_s/2$ of a sinusoid that yields samples identical to those obtained in part (c)?

(e) Same as part (c) except $F_s = 20$ Hz.

5. Aliasing and Sampling

Solution:

part (a): Given analog signal can be written as:

$$x_a(t) = x(t) = 3\cos(2\pi \times 50 \times t) \quad \text{— (i)}$$

Recall!

$$x(t) = A\cos(2\pi \times F \times t) \quad \text{— (ii)}$$

Comparing (i) & (ii) we get:

$$F = 50\,Hz$$

or

$$F_{max} = 50\,Hz$$

According to Nyquist criteria $F_s = F_{s(min)} = 2F_{max}$

$$F_{s(min)} = 2 \times 50\,Hz = 100\,Hz$$

Comments:

$F_{s(min)}$ should be atleast 100 Hz to avoid aliasing.

5. Aliasing and Sampling

part(b)

Recall!

$$x(n) = A\cos(2\pi f n + \phi).$$

where:

$$f = \frac{F}{F_s} = \frac{F_{max}}{F_s} = \frac{50}{200} = \frac{1}{4} = 0.25 < 0.5$$

(Condition satisfied)

Therefore:

$$x(n) = A\cos\left(2\pi\left(\frac{1}{\cancel{4}\,2}\right)\right)n$$

$$x(n) = A\cos\left(\frac{\pi}{2}\right)n = 3\cos\left(\frac{\pi}{2}\right)n.$$

5. Aliasing and Sampling

part(c)

$$f = \frac{F_{max}}{F_s} = \frac{50}{75} = \frac{2}{3} = 0.67 > 0.5$$ (condition isn't satisfied)

Therefore we need to normalize 'f' so that it must be bounded b/w -0.5 to 0.5

$$f = \frac{2}{3} \quad \text{— (i)}$$

$$f = 1 \pm \frac{x}{3} \quad \text{— (ii)}$$

$$\frac{2}{3} = 1 \pm \frac{x}{3}$$

$$-1 = \pm x \; ; \quad x = -1 \;\; \& \;\; -x = -1 , \; x = 1$$

for $x = 1$

$$f = 1 + \frac{x}{3} = \frac{4}{3} = \text{not same as (i)}$$

for $x = -1$

$$f = 1 - \frac{1}{3} = \frac{2}{3} = \text{same as (i)}$$

5. Aliasing and Sampling

Now $x(n) = A\cos(2\pi f n + \phi)$

$$= A\cos\left(2\pi\left(1 - \frac{1}{3}\right)n\right)$$

$$= 3\cos\left(2\pi - \frac{2\pi}{3}\right)n$$

$$x(n) = -3\cos\left(\frac{2\pi}{3}\right)n$$

5. Aliasing and Sampling

part(d):

$F = ?$

$f = \frac{F}{F_s}$

$F = f \times F_s$ —(i)

in part (c) $F_s = 75\,Hz$ and $x(n) = 3\cos\left(\frac{2\pi}{3}\right)n$

compare $x(n)$ with:

$x'(n) = A\cos(2\pi f n + \phi)$

$x(n) = 3\cos\left(2\pi \times \frac{1}{3}\right)n$

$x'(n) = A\cos\left(2\pi \times f\right)n$

we get $f = \frac{1}{3}$

Now eq (i) $F = \frac{1}{3} \times 75 = 25\,Hz$

& $x(t) = 3\cos(2\pi F t + \phi)$

$= 3\cos\left(2\pi(25)t\right)$

$x(t) = 3\cos(50\pi t)$

5. Aliasing and Sampling

part (e):

$$f = \frac{F_{max}}{F_s} = \frac{F}{F_s} = \frac{50\ Hz}{20Hz} = \frac{5}{2} = 2.5 > 0.5$$ (condition not satisfied)

Therefore we need to normalize 'f' such that it must be bounded b/w −0.5 to 0.5.

$$f = \frac{5}{2} \quad \text{--(i)}$$

$$f = 1 \pm \frac{x}{2} \quad \text{--(ii)}$$

$$\frac{5}{2} = 1 \pm \frac{x}{2}$$

$3 = \pm x$; $x = -3$ & $x = 3$

5. Aliasing and Sampling

for $x = -3$

(ii) $\Rightarrow f = 1 - \frac{3}{2} = -\frac{1}{2}$ not same as i

for $x = 3$

(ii) $\Rightarrow f = 1 + \frac{3}{2} = \frac{5}{2}$ = same as (i)

we need further normalization of 'f' such that it must be bounded b/w -0.5 & 0.5

$$f'' = \frac{3}{2} \quad \text{— (iii)}$$

$$f'' = 1 \pm \frac{x}{2} \quad \text{— (iv)}$$

$$\frac{3}{2} = 1 + \frac{x}{2}$$

$$1 = \pm x$$

for $x = 1$

$f'' = 1 + \frac{1}{2}$ = same as (iii)

$\frac{1}{2} \rightarrow 0.5$, condition satisfied

Therefore:

$$x(n) = A\cos(2\pi f'' n + \phi).$$

$$= A\cos\left(2\pi\left(1 + \frac{1}{2}\right)n\right) = A\cos\left(2\pi + \frac{2\pi}{2}\right)n.$$

$$x(n) = 3\cos\left(\frac{2\pi}{2}\right)n = 3\cos(\pi)n.$$

5. Aliasing and Sampling

Example 3:

Consider the following analog signal:

$$x(t) = 25\cos 50\pi t + 50\sin 100\pi t + 75\cos 150\pi t$$

Determine:

(a) Discrete time signal and plot it for n=0, 1, 2 & 3

(b) Minimum sampling rate to avoid aliasing.

(c) Minimum and Maximum Analog & digital frequencies.

(d) Normalized frequencies

5. Aliasing and Sampling

Solution:

part(a): $x(t)$ can be written as:

$$x(t) = 25\cos(2\pi \times \overset{F_1}{25} \times t) + 50\sin(2\pi \times \overset{F_2}{50} \times t) + 75\cos(2\pi \times \overset{F_3}{75} \times t)$$

$$F_s = 2F_{max} = 2 \times F_3 = 2 \times 75Hz = 150Hz.$$

Now:

$$f_1 = \frac{F_1}{F_s} = \frac{25Hz}{150Hz} = \frac{1}{6}Hz = 0.17Hz$$

$$f_2 = \frac{F_2}{F_s} = \frac{50Hz}{150Hz} = \frac{1}{3}Hz = 0.33Hz$$

$$f_3 = \frac{F_3}{F_s} = \frac{75Hz}{150Hz} = \frac{1}{2}Hz = 0.5Hz$$

$\Rightarrow$ f_1, f_2 & f_3 are bounded b/w -0.5 to 0.5

Therefore D.T signal can be written as:

$$x(n) = 25\cos(2\pi f_1 n) + 50\sin(2\pi f_2 n) + 75\cos(2\pi f_3 n)$$

$$\boxed{x(n) = 25\cos(2\pi(0.17)n) + 50\sin(2\pi(0.33)n) + 75\cos(2\pi(0.5)n)}$$

5. Aliasing and Sampling

Plot of a Signal:

for $n=0$; consider $\pi = 3.14$ radian.

$x(0) = 25\cos(2\times 3.14\times 0.17\times 0) + 50\sin(2\times 3.14\times 0.33\times 0) + 75\cos(2\times 3.14\times 0.5\times 0)$

$\boxed{x(0) = 100}$

for $n=1$;

$x(1) = 25\cos(2\times 3.14\times 0.17\times 1) + 50\sin(2\times 3.14\times 0.33\times 1) + 75\cos(2\times 3.14\times 0.5\times 1)$

$x(1) = 101.6$

$\boxed{x(1) \simeq 102}$

for $n=2$;

$x(2) = 103.1$

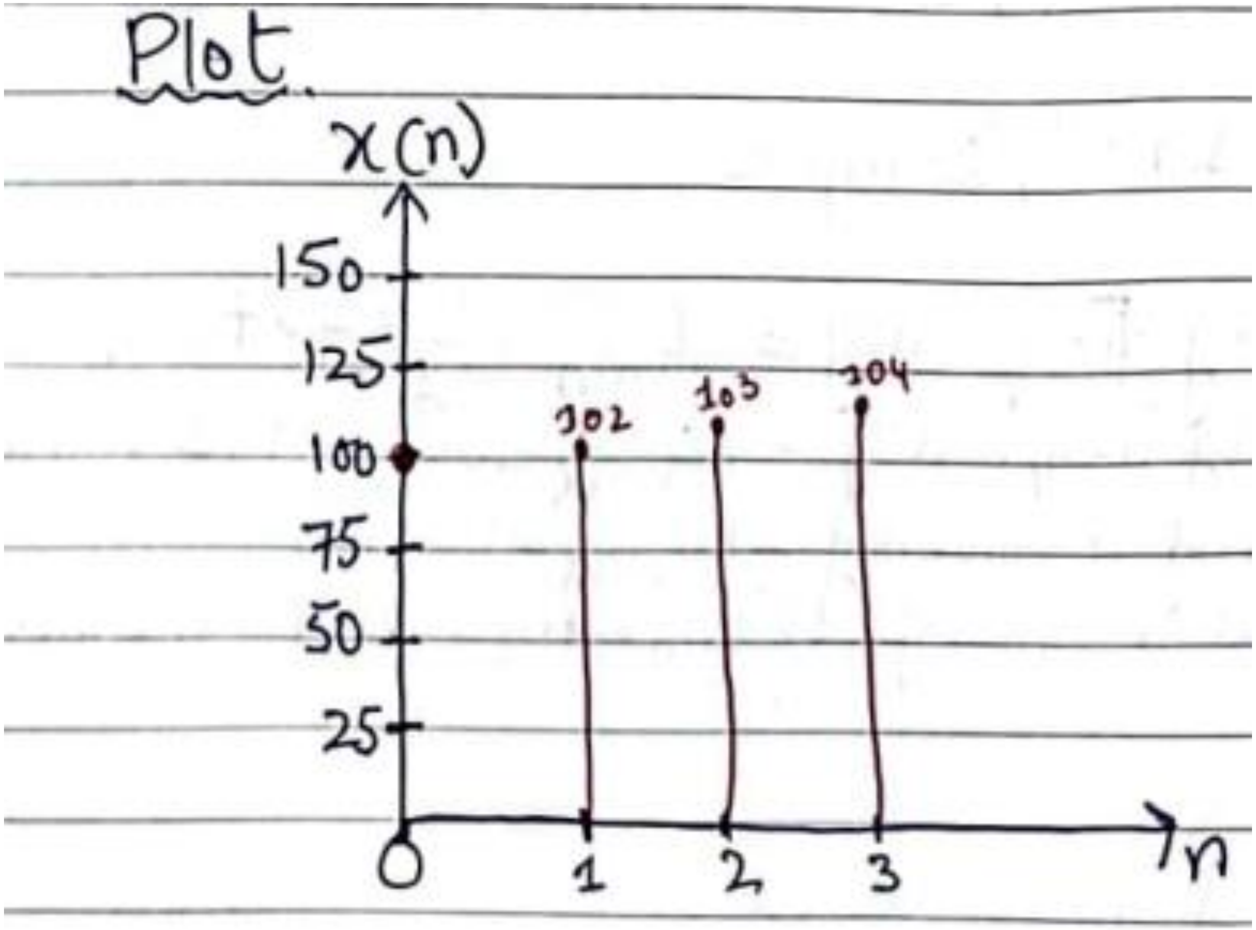
$\boxed{x(2) \simeq 103}$

for $n=3$;

$x(3) = 104.3$

$\boxed{x(3) \simeq 104}$

5. Aliasing and Sampling



5. Aliasing and Sampling

part(b):

$$F_{s(min)} = 2F_{max} = 2(75) = 150\ Hz.$$

part(c):

Analog Frequency $= \Omega = 2\pi F$ (in rad/sec)

Therefore:

$$\Omega_1 = 2\pi F_1 = 2 \times 3.14 \times 25Hz = 157\ rad/sec$$

$$\Omega_2 = 2\pi F_2 = 2 \times 3.14 \times 50Hz = 314\ rad/sec$$

$$\Omega_3 = 2\pi F_3 = 2 \times 3.14 \times 75Hz = 471\ rad/sec.$$

5. Aliasing and Sampling

Digital Frequency = $\omega = \Omega T$ (in rad/sample)

Therefore:

$\therefore T = \frac{1}{F_s} = \frac{1}{150} = 6.6\,\text{msec}$

$\omega_1 = \Omega_1 T = 157 \times 6.6\,\text{msec}.$

$\omega_1 = 1.04$ rad/sample

$\omega_2 = \Omega_2 T = 314 \times 6.6\,\text{msec}$

$\omega_2 = 2.07$ rad/sample

$\omega_3 = \Omega_3 T = 471 \times 6.6\,\text{msec}$

$\omega_3 = 3.11$ rad/sample.

5. Aliasing and Sampling

* Minimum Analog frequency $= \Omega_{min} = \Omega_1 = 157$ rad/sec
* Maximum Analog Frequency $= \Omega_{max} = \Omega_3 = 471$ rad/sec
* Maximum Digital Frequency $= \omega_{max} = \omega_3 = 3.11$ rad/sample
* Minimum Digital Frequency $= \omega_{min} = \omega_1 = 1.04$ rad/sample.

part (d):

Normalized frequencies:

$f_1 = \frac{F_1}{F_s} = 0.17\,Hz$ similary $f_2 = 0.33\,Hz$ & $f_3 = 0.5\,Hz$.

6. Quantization and Coding

① **Quantization:** It converts a signal continuous in amplitude into a signal discrete in amplitude.

② **Quantization Error:** The error introduced in representing continuous-value signal by a finite set of discrete value levels is called Quantization Error or Quantization Noise.

> Continuous Valued Signal:
If the signal takes infinite set of values it is called continuous valued signal.

> Discrete Valued Signal:
If the signal takes finite set of values it is called discrete valued signal.

Quantization Error = $e_q(n) = x_q(n) - x(n)$

6. Quantization and Coding

③ **Range of Quantization Error:**

$$-\frac{\Delta}{2} \leq e_q(n) \leq \frac{\Delta}{2}$$

④ **Resolution or Step size:**

The distance Δ b/w two successive quantization levels is called step size or Resolution.

⑤ **Quantization levels:**

The values allowed in the digital signal are called called the Quantization levels. Number of Quantization Levels = L. For each iteration of 'n' results in one quantization level. For example for n=0 to n=5 means that there are '6' quantization Levels.

6. Quantization and Coding

⑥ Uniform Quantization: Quantization step size is same or fixed for a complete signal. We will consider uniform quantization for our understanding purpose.

⑦ Dynamic Range: It is the difference between x_{max} and x_{min}.

⑧ Signal-to-quantization-noise-ration (SQNR or SN_qR):

It is the ratio of the signal power to the noise power. SQNR is expressed in decibel as:

$$SQNR(dB) = 1.76 + 6.02b$$

where b = number of bits used for each code for each level.

⑨ Coding: In coding process, each discrete value is represented by b-bit binary sequence.

$$2^b \geq L$$

Also

$$b \geq \log_2(L)$$

6. Quantization and Coding

- Example: For the given discrete time signal

$$x(n) = \begin{cases} 0.9^n & n \geq 0 \\ 0 & n < 0 \end{cases}$$

a) Quantization Error.

b) Resolution.

c) Range of Quantization Error.

d) Digital Signal.

6. Quantization and Coding

Solution: (3)

n	$x(n)$ D.T signal	$x_q(n)$ (Truncation) [1-Digit Truncation]	$x_q(n)$ (Rounding) [1-Digit Rounding]	$e_q(n) = x_q(n) - x(n)$ (Rounding) We have consider $x_q(n)$ as rounding.
0	$= 0.9^n = 0.9^0 = 1$	1	1	$e_q(0) = x_q(0) - x(0) = 1 - 1 = 0$
1	$= 0.9^n = 0.9^1 = 0.9$	0.9	0.9	$e_q(1) = x_q(1) - x(1) = 0.9 - 0.9 = 0$
2	$= 0.9^n = 0.9^2 = 0.81$	0.8	0.8	$e_q(2) = x_q(2) - x(2) = 0.8 - 0.81 = -0.01$
3	0.729	0.7	0.7	−0.029
4	0.6561	0.6	0.7	0.0439
5	0.59049	0.5	0.6	0.00951
6	0.531441	0.5	0.5	−0.031441
7	0.4782969	0.4	0.5	0.0217031
8	0.43046721	0.4	0.4	−0.03046721
9	0.387420489	0.3	0.4	0.012579511
10	0.34867844	0.3	0.3	0.04867844
11	0.313810596	0.3	0.3	0.013810596
12	0.282429536	0.2	0.3	0.017570463

6. Quantization and Coding

a) solution:

Refer to column 5 of Table.

b) Resolution or stepsize:

Suppose resolution between quantization level 4 and 5. Also assume $x_q(n)$ (rounding)

$$\Delta = x_q(4) - x_q(5)$$

$$= 0.7 - 0.6$$

$$\boxed{\Delta = 0.1}$$

c) Range of Quantization Error:

It is given as:

$$-\frac{\Delta}{2} \leq e_q(n) \leq \frac{\Delta}{2}$$

$$-\frac{0.1}{2} \leq e_q(n) \leq \frac{0.1}{2}$$

$$-0.05 \leq e_q(n) \leq 0.05 \Rightarrow$$ (Condition satisfied Refer to Column 5 of Table)

6. Quantization and Coding

d) Conversion of given signal '$x(n)$' to digital signal

We know that:

$$2^b \geqslant L$$

and

$$b \geqslant \log_2(L)$$

or

$$b = \log_2(L)$$

from table we can determine $L = 13$, Therefore:

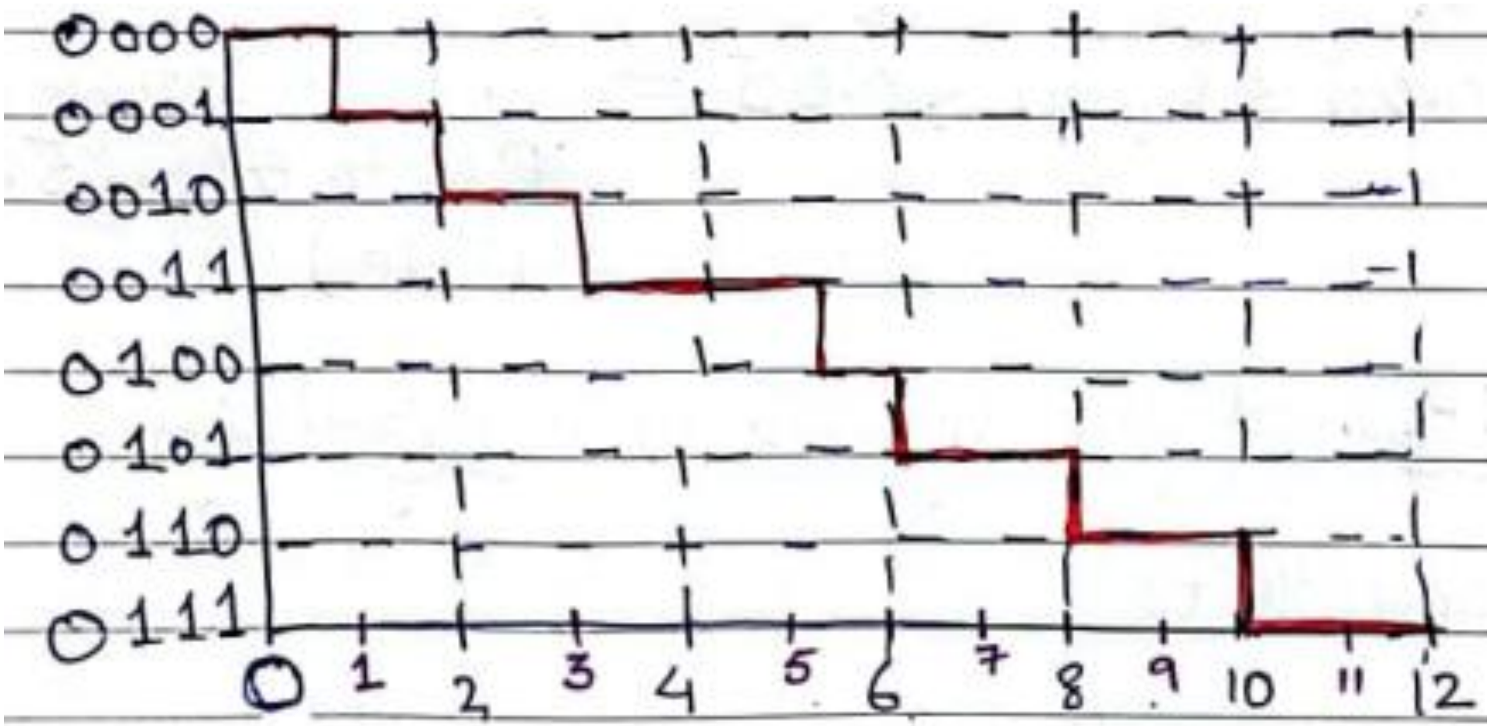
$$b = \log_2(13) = 3.7 \cong 4$$

6. Quantization and Coding

Coding:

n	$x_q(n)$ (Round off)	Code	
0	1	0 0 0 0	0
1	0.9	0 0 0 1	1
2	0.8	0 0 1 0	2
3	0.7	0 0 1 1	3
4	0.7	0 0 1 1	
5	0.6	0 1 0 0	4
6	0.5	0 1 0 1	5
7	0.5	0 1 0 1	
8	0.4	0 1 1 0	6
9	0.4	0 1 1 0	
10	0.3	0 1 1 1	7
11	0.3	0 1 1 1	
12	0.3	0 1 1 1	

6. Quantization and Coding



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2. *“Digital Signal Processing Using Matlab”, 4th Edition by Vinay K. Ingle, John G. Proakis, (2016), Cengage Learning.*
3. *Related material from open source, mainly internet.*